The examination is being carried out on the **following application documents**

Description, Pages

1-106 as published

Claims, Numbers

1-15 filed with entry into the regional phase before the EPO

Drawings, Sheets

1/20-20/20 as published

1 The following documents have been cited in the international search report; the numbering will be adhered to in the rest of the procedure.

D1 WO 2018/022933 A1 (OBI PHARMA, INC, ET AL.) 1 February 2018 (2018-02-01)

D2 WO 2017/185089 A2 (OBI PHARMA, INC., ET AL.) 26 October 2017 (2017-10-26)

D3 WO 2017/004150 A1 (THE JOHNS HOPKINS UNIVERSITY, ET AL.) 5 January 2017 (2017-01-05)

2 Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D4 DANYANG FENG ET AL: "Recent Advance in Tumor-associated Carbohydrate Antigens (TACAs)-based Antitumor Vaccines",ACS CHEMICAL BIOLOGY, vol. 11, no. 4, 1 January 2016 (2016-01-01), pages 850-863, XP055428997,ISSN: 1554-8929, DOI: 10.1021/acschembio.6b00084

D5 JIN KE-TAO ET AL: "Recent advances in carbohydrate-based cancer vaccines", BIOTECHNOLOGY LETTERS, KLUWER ACADEMIC PUBLISHERS, DORDRECHT, vol. 41, no. 6, 16 April 2019 (2019-04-16), pages 641-650, XP036798893, ISSN: 0141-5492, DOI: 10.1007/S10529-019-02675-5 [retrieved on 2019-04-16]

D6 SCHIJNS VIRGIL ET AL: "Modulation of immune responses using adjuvants to facilitate therapeutic vaccination",IMMUNOLOGICAL REVIEWS, vol. 296, no. 1, 28 June 2020 (2020-06-28), pages 169-190, XP055824532,US ISSN: 0105-2896, DOI: 10.1111/imr.12889Retrieved from the Internet:URL:https://onlinelibrary.wiley.com/doi/full-xml/10.1111/imr.12889

D7 WO 2020/198699 A2 (OBI PHARMA INC; LAI MING TAIN) 1 October 2020 (2020-10-01)

3 Art. 123(2) EPC

The Applicants are invited to indicate on which parts of the application as originally filed the new claims are based. Claim 1 as originally filed does not mention an adjuvant, does not refer to prevention (and/or treatment either) and only discloses advanced gastric cancer and not gastric cancer only. For all these reasons, claims 1-10 on file are objected to under Art. 123(2) EPC.

4 Art. 83 EPC

In order to fulfill the requirements of Article 83 EPC the sufficiency of disclosure must be satisfied at the effective date of the patent, i.e. on the basis of the information in the patent application together with common general knowledge then available to the skilled person (cf. T609/02 point 8 and 13 of the reasons). Where a therapeutic application is claimed in the form of the use of a substance or composition for a defined therapeutic application, attaining the claimed therapeutic effect is a functional feature of the claim. As a consequence, under Article 83 EPC the application must disclose the suitability of the product to be manufactured for the claimed therapeutic application (see G2/88 and G6/88, Headnote III. and point 9 of the reasons;

T609/02, point 2). Since this is not the case for the administration schedule falling under the scope of claims 11 (a)-(d), the requirements of Art. 83 EPC are not considered to be met. The only real schedule of administration actually disclosed can be found in the examples 1-3 and it concerns a single disease only using very specific therapeutics. In that context, it is noted that none of the supplementary or optional features of claim 8 would appear to have been assessed by the Applicants in the present application.

5 Art.84.EPC

5.1 Abbreviations for biological material are often less well known than the Applicants assume and should therefore be avoided (see EPO Guidelines, F-III, 6.3). This objection concerns the features OBI-821 and OBI-833.

Document WO2019/191317 (cited in description) discloses according to the present application OBI-821. However, this document discloses several variants (see for instance fig. 3A), leading to doubts as to the scope of this feature.

Document WO2014/107652 (cited in description) does not seem to disclose OBI-833 at all, but only OPT-821.

The scope of claims 5, 7 and 11-15, which should be self-contained (EPO Guidelines, F-IV, 4.1), is thus currently considered to be unclear. The requirements of Art. 123(2) EPC should be strictly observed to overcome these defects.

5.2 Claims 12 and 13 refer to standard immune parameters which are inherent to each and every vaccine combining a carrier and an antigen. They thus do not add anything specific to claims 12 and 13 on file.

5.3 Claims 6 and 15 are not clear, since they encompass the option of having no antigen at all. This is in contradiction with claims 1 and 11 which relate to therapeutically active entities which require the presence of an antigen.

6 Novelty, Art. 54 EPC

In view of the objections raised under Art.123(2) EPC and Art. 84 EPC above, the analysis below must be considered to be preliminary and may have to be reconsidered based on the Applicants counter-arguments.

6.1 D1 discloses the treatment of mice with SSEA-4-KLH vaccine or SSEA-4-DT + OBI-821, KLH was a better carrier than DT [0138-139]. Similarly, a bivalent vaccine containing SSEA-4-KLH and Globo H-KLH with OBI-821 adjuvant was also tested [0141-0142]. Finally, a trivalent SSEA-4-KLH + Globo H-KLH +SSEA-3-KLH was combined with OBI-821 [0144-0145]. All the vaccines tested led to an immune response and are thus efficient immunogens. Stomach, lung, colon and breast cancer are potential targets. Combinations with chemotherapy and cytotoxic agents are also disclosed [004, 006, 020, 024-029, 059, 075-077, 095, 096].

Claims 1-15 are thus not considered to be novel over D1.

6.2 D2 discloses a Globo H-KLH conjugate used in combination with OBI-821, in combination with low dose cyclophosphamide against metastatic breast cancer [098]. Further, the same treatment was used against ovarian cancer (example 2).

Claims 1-3, 5 and 8-10 are thus not considered to be novel over D2.

6.3 D4 discloses the use of Globo H (GH) as vaccine against breast cancer, in combination with KLH, DT CRM197, TT or BSA as carrier, and an adjuvant being alpha-galactosylceramide C34. The highest antibody titers were obtained with GH-DT/C34. These antibodies bind to GH, SSEA-3 and SSEA-4. This document also discloses QS21 in combination with five prostate and breast cancer associated carbohydrates, Globo-H, GM2, sTn, TF and Tn coupled to KLH. This led to strong antibody responses (p. 852-854)

Claims 1-7, 9 and 10 are thus not considered to be novel over D4.

7 Inventive step, Art. 56 EPC

D5 lists on page 643, fig. 1 some cancers expressing Globo-H, SSEA-3 or SSEA-4, including lung cancer. D6 further links small-cell lung cancer with Globo-H (p. 76, Table 2). As to D7, it highlights on page 7, figs. 2A-2E and 12, that NSCLC specimen express Globo-H, making it an obvious target for treating this specific disease, as well as all other diseases mentioned in the same context in D7, see p. 7-9.

In view of this, it is not apparent to the division on the basis of which combination of features an inventive step could be acknowledged for independent claims 1, 11 or 14, assuming that novelty could be eventually be acknowledged.

8 In order to facilitate the examination of the conformity of the amended application with the requirements of Article 123(2) EPC, the applicant is requested to clearly identify the amendments carried out, irrespective of whether they concern amendments by addition, replacement or deletion, and to indicate the passages of the application as filed on which these amendments are based.